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Baker College Teacher Prep Lesson Plan Format

Subject Area & Grade Level: Mathematics – Grade 6

Lesson Duration: 60 minutes

<u>Lesson Goal:</u> What do we want students to learn?	<u>Assessment:</u> How will we know if they have learned it? (see guide)	<u>Intervention:</u> What will we do if they don't learn it? (see guide)	<u>Enrichment:</u> What will we do if they already know it? (see guide)
What do we want students to learn? Students will apply and extend previous understandings of numbers to the system of rational numbers by identifying, comparing, and ordering positive and negative numbers on a number line and coordinate plane, and by understanding the concept of absolute value in real-world contexts. This aligns with the Michigan Department of Education (n.d.) Grade 6 Number System standards found on page 45, under the section Apply and extend previous understandings of numbers to the system of rational numbers through.	☒ Formative – During the lesson, students will complete an exit ticket in which they place a set of rational numbers, such as positive, negative, fractions, and decimals, on a number line, write an inequality statement, and explain the absolute value of a given number in a real-world scenario. This could be a bank account balance of -\$45. The teacher will circulate during partner work and use Think/Pair/Share to check for understanding throughout the lesson.	☒ Manipulatives – Students who have not mastered the concept will use physical number line strips and integer tiles to build concrete understanding of positive and negative values. ☒ Small group instruction – The teacher will pull a small group during the enrichment/independent practice portion of the lesson to re-teach the concept of absolute value using a different approach, which will be distance from zero using a classroom floor number line. ☒ Re-teach in a different way – Instead of abstract notation, students will use color-coded cards, red for negative, blue for positive, to physically order numbers before transitioning to the number line diagram. ☒ Graphic organizers – A structured notes	☒ Centers – High level – Students who demonstrate mastery early will move to an enrichment center where they solve real-world coordinate plane problems, such as plotting and finding distances between cities on a map grid. ☒ Independent projects – Advanced students will create their own illustrated real-world rational number scenario book that shows at least five examples of positive/negative numbers in context, such as temperature, elevation, finance, sports yardage, sea depth. ☒ Peer tutoring – Mastery students may serve as Math Mentors and assist a partner at the same table during guided practice, explaining their

		template will be provided that includes a number line, a Frayer model for absolute value, and sentence stems for writing inequality statements.	reasoning using academic vocabulary. ☒ Games – Students may engage with a rational number ordering card game, racing to correctly sequence a hand of rational number cards on their personal number line.
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Common Core State
Standard on page 45
from source Michigan
Department of
Education (n.d.):

The following parts from the
section Apply and extend
previous understandings of
numbers to the system of
rational numbers and the
section itself.

☐ 6.NS.5 – Understand that
positive and negative numbers
describe quantities having
opposite directions or values
(e.g., temperature above/below
zero, elevation above/below
sea level, credits/debits,
positive/negative electric
charge); use positive and
negative numbers to represent
quantities in real-world
contexts, explaining the
meaning of 0 in each situation.

☐ 6.NS.6 – Understand a
rational number as a point on
the number line. Extend
number line diagrams and
coordinate axes familiar from
previous grades to represent
points on the line and in the
plane with negative number
coordinates.

- 6.NS.6a – Recognize
opposite signs of
numbers as indicating
locations on opposite
sides of 0 on the
number line;
recognize that the
opposite of the
opposite of a number
is the number itself,
e.g., $-(-3) = 3$, and
that 0 is its own
opposite.
- 6.NS.6c – Find and
position integers and
other rational
numbers on a
horizontal or vertical
number line diagram;
find and position
pairs of integers and
other rational
numbers on a
coordinate plane.

☐ 6.NS.7 – Understand
ordering and absolute value of
rational numbers.

• 6.NS.7a – Interpret <u>statements of inequality as statements about the relative position of two numbers on a number line diagram.</u> • 6.NS.7c – Understand <u>the absolute value of a rational number as its distance from 0 on the number line; interpret absolute value as magnitude for a positive or negative quantity in a real-world situation.</u> <u>Learning Objective:</u> (In words that your students would understand) <u>I can place positive and negative numbers on a number line, compare them using greater than and less than, and explain what absolute value means using a real-life example like money or temperature.</u>			
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Materials:

- ❑ Number line strips (physical, laminated)
- ❑ Integer tiles (red and blue)
- ❑ Dry-erase markers and personal whiteboards
- ❑ Coordinate plane graph paper
- ❑ Exit ticket worksheet
- ❑ Projector/Smartboard
- ❑ *Mathematical Mindsets* 2nd ed. by Jo Boaler Boaler, 2022 – referenced for growth mindset framing during lesson introduction
- ❑ Michigan Math Standards document by the Michigan Department of Education (n.d.)

Select your learning strategy:

- Direct Teach

- Demonstration
- Cooperative Learning
- Discovery/Inquiry-Based Learning
- Project-Based Learning
- Reading/Writing/Math Workshop
- Other

? ☒ Cooperative Learning

? ☒ Demonstration

? ☒ Discovery/Inquiry-Based Learning

Activities Planned: ☐ Active (Students are active participants in learning)

☐ Passive (Teacher led lecture/demonstration)

☒ Both

Lesson Delivery Steps:

Step 1 – Hook/Introduction (5 min): The teacher opens with a real-world scenario. Imagine you owe your friend \$15 but you have \$8 in your wallet. Which is bigger — the amount you owe or the amount you have? Students turn and talk. Teacher introduces the idea of negative numbers as representing what we owe or situations below zero. Thus, connecting to the standard from Michigan Department of Education (n.d.) on page 45, which is apply and extend previous understandings of numbers to the system of rational numbers.

Step 2 – Direct Instruction/Modeling (10 min): Teacher models placing integers and rational numbers on a number line on the Smartboard. Teacher explicitly models how to determine which number is greater using position on the number line, which supports 6.NS.7a, and models finding absolute value as distance from zero, which supports 6.NS.7c.

Step 3 – Guided Practice (15 min): Students work in pairs using laminated number line strips and integer tiles. They receive a card set of rational numbers and must place them in order, then write two inequality statements. Teacher circulates and uses targeted questioning.

Step 4 – Cooperative Activity (15 min): Small groups of 3–4 students plot a set of coordinate pairs in all four quadrants on graph paper and identify points that are reflections across axes, which supports 6.NS.6. Groups share out one finding.

Step 5 – Exit Ticket (10 min): Students independently complete the exit ticket: place 5 numbers on a number line, write an inequality, and explain the absolute value of -30 in the context of a bank account.

Step 6 – Closure (5 min): Teacher facilitates a brief class discussion asking students to share one thing that surprised them and one connection they made to real life. This type of open reflective discussion supports the broader principle, described by Boaler (2022) in Chapter 8 from section Assessment for a Growth Mindset and the border idea of schools having a race to now where, that moving away from performance-only feedback and toward diagnostic, discussion-based learning deepens student understanding and builds positive mathematical identity.

Check the Core Teaching Practices addressed in your lesson: (from MDE CTPs)

- X Leading a group discussion (CTP #1)
- X Explaining and modeling content, practices, and strategies (CTP #2)
- X Eliciting and interpreting individual students' thinking (CTP #3)
- X Building respectful relationships with students (CTP #10)
- X Check for Understanding (CTP #15)

List real-world connections including attention to English language learners and culturally and historically responsive practices (diversity, inclusion, equity, and social justice):

Real-world contexts include bank account balances, weather temperatures, football yardage, ocean depth, and elevation — scenarios accessible to diverse learners. For English Language Learners, sentence stems are provided.

Ex:

"The absolute value of ____ is ____ because ____"

The lesson acknowledges that different cultures use number systems differently, and validates varied lived experiences through examples, such as discussing temperatures in Celsius vs. Fahrenheit, or elevations relevant to geography familiar to students' home countries.

List technology tools:

- ☐ Smartboard / Projector for number line modeling
- ☐ Optional: Desmos online graphing tool for coordinate plane activity

List collaboration opportunities (whole group, small group, partnerships, building resource personnel i.e., school social worker, special educators, parents, etc.):

- ☐ Partner work during guided practice
- ☐ Small group work during cooperative activity
- ☐ Peer tutoring for enrichment students
- ☐ Building resource personnel: special educator may co-teach or support during small group intervention

Lesson Plan Guide

Assessment: Used to gather information about a student's progress towards mastery of the learning objective, help the teacher identify what instruction is working well and what needs refinement, and informs the students about their learning.

Options to consider

- ☐ Diagnostic/Pre-Assessment – Used to check prior knowledge **before** a lesson
- ☐ Self-Assessment (Writing Prompts, Running Records, Performance Task, Other)
- ☒ Formative – Used **during** a lesson to check progress, identify any misconception, and give feedback to students (Learning/Response Log, Admin/Exit Ticket, Think/Pair/Share, One Minute Paper, Other)
- ☐ Summative – Used at the **end** of a lesson to check student mastery of the objective (End of Unit Test, Final Exams or Mid-term Exams, State Tests, Culminating Project, Portfolio, Other)

Student-Created Rubric: Rational Numbers on the Number Line – Exit Ticket

Criteria	4 – Exceeds Expectations	3 – Meets Expectations	2 – Partially Meets Expectations	1 – Does Not Meet Expectations
Number Line Placement (6.NS.6c)	All 5 rational numbers are placed correctly and precisely on the number line, including fractions and negatives.	4 of 5 numbers are placed correctly with minor spacing errors.	2–3 numbers are placed correctly; significant errors with negatives or fractions.	Fewer than 2 numbers are placed correctly; little to no understanding of rational number placement shown.
Inequality Statement (6.NS.7a)	Two inequality statements are written correctly using $<$ or $>$ with accurate interpretation of position on the number line.	One correct inequality statement is written; a second is attempted but contains a minor error.	One inequality statement is attempted but contains errors in symbol use or number comparison.	No inequality statement is written, or statements are entirely incorrect.
Absolute Value Explanation (6.NS.7c)	Absolute value is defined accurately as distance from zero AND is correctly applied to the bank account context with a clear, complete explanation.	Absolute value is correctly calculated and a basic real-world explanation is given, though not fully developed.	Absolute value is calculated correctly but no real-world connection is made, or the explanation is unclear.	Absolute value is not calculated correctly and/or no explanation is provided.

Criteria	4 – Exceeds Expectations	3 – Meets Expectations	2 – Partially Meets Expectations	1 – Does Not Meet Expectations
Mathematical Vocabulary	Three or more academic terms (e.g., rational number, absolute value, integer, opposite, inequality) are used accurately throughout the exit ticket.	Two academic terms are used accurately.	One academic term is used; others are missing or used incorrectly.	No academic vocabulary is used or all terms are used incorrectly.
Real-World Connection (6.NS.5)	The student clearly connects the absolute value of -30 to a real-world banking context with a complete sentence and accurate mathematical reasoning.	The student connects -30 to the banking context with a mostly complete explanation.	The student references the context but the mathematical reasoning is incomplete or partially incorrect.	No real-world connection is made or the connection is entirely inaccurate.

Analysis: The Good, the Bad, and the Ugly

The Good (Positives)

- The rubric is directly aligned with each Michigan state standard addressed in the lesson, making it easy to connect scores back to specific learning objectives. This alignment ensures that the assessment is purposeful rather than general.
- Each criterion uses precise, measurable descriptors. For example, all 5 rational numbers are placed correctly, which leaves no ambiguity. The teacher can score it without subjective judgment.
- The rubric simultaneously serves as feedback for students and as a diagnostic tool for the teacher. A student who scores 1 on Absolute Value Explanation but 3 on Number Line Placement immediately reveals a misconception pattern the teacher can target in small group intervention. Boaler (2022) explains in Chapter 8, *Assessment for a Growth Mindset*, that diagnostic comments — rather than grades alone — cause students to learn at significantly faster rates and close achievement gaps between student groups.
- The inclusion of a Mathematical Vocabulary criterion supports language development for English Language Learners, which aligns with the culturally responsive goals of the lesson.

The Bad (Challenges)

- The five-criteria rubric could be time-consuming to score for every student during a single exit ticket. A teacher with 30 students may find this rubric burdensome to use consistently across the whole class.
- The Partially Meets Expectations descriptors for some criteria rely on quantity — for example, counting how many numbers are placed correctly — rather than the quality of the student's reasoning. This may not fully capture why a student is struggling or which specific misconception is driving the error.
- The rubric does not account for the process or strategies students used, only the accuracy of their final product. Boaler (2022) argues in Chapter 8, *Assessment for a Growth Mindset*, from the discussion of what good assessment should measure, that tests and rubrics in mathematics have historically focused on what is easy to measure rather than on the deep, creative, and analytical thinking that truly matters. A future rubric should include a row for mathematical reasoning and work shown, honoring student thinking alongside correct answers.

The Ugly (Visual Appearance)

- In table format, the rubric is text-heavy and may be difficult for 6th graders to read independently. The dense language in each cell could overwhelm students who struggle with reading, particularly English Language Learners or students with IEPs. A student-facing version should use simplified language and perhaps icons or color-coding.
- The four-point scale labels — Exceeds, Meets, Partially Meets, Does Not Meet — are useful for teacher scoring but may feel discouraging for students who see the lowest category repeated across multiple criteria. Boaler (2022) describes in Chapter 8, *Assessment for a Growth Mindset*, in the introduction from the 9th paragraph on the portion on grading and its alternatives, that when students receive grades or judgment-based feedback, they tend to define themselves by those labels rather than using them as information to grow. Replacing evaluative labels with growth-oriented language such as *Getting Started*, *On My Way*, *Almost There*, and *Got It* would better support a growth mindset classroom culture.
- The table itself, without color, borders, or visual hierarchy, is monotonous. Adding shading or color to distinguish performance levels — for example, green for the highest and a warm tone for the lowest — would make the rubric more visually intuitive and practical for daily classroom use.

Reflection Paragraph

Creating this rubric was an illuminating process that revealed both the strengths and limitations of my current approach to assessment design. First, I recognize that rubric alignment to specific standards is a non-negotiable starting point — without it, scoring becomes subjective and instructional follow-through becomes disconnected from student data. Moving forward, I will always draft rubric criteria directly from the lesson standard language before expanding into performance descriptors, ensuring every row traces back to a measurable objective. Second, this activity highlighted that rubrics must serve two audiences simultaneously: the teacher scoring work and the student using it to self-assess. My current rubric is teacher-facing in its complexity and language. Boaler (2022) explains in Chapter 8, *Assessment for a Growth Mindset*, in the sections about discussing the effects of grading on student identity, such as in the introduction form paragraph 9, that when students receive frequent evaluative feedback, they begin to see themselves as that score rather than as a growing learner. In future practice, I will create parallel student-friendly versions using simplified vocabulary and growth-oriented labels so that the rubric becomes a motivational tool, not just a measurement tool. Third, I will intentionally add a mathematical reasoning criterion to future rubrics so that student thinking and strategy are recognized alongside correct final answers. Boaler (2022) reinforces throughout Chapter 8, *Assessment for a Growth Mindset*, that the most powerful assessments communicate diagnostic information about what a student understands and what they need next — not simply whether they got it right. Embedding process into my rubric design will allow me to identify and address misconceptions earlier and more precisely, which is ultimately the greatest purpose of any classroom assessment tool.

References

Boaler, J. (2022). *Mathematical mindsets* (2nd ed.). Jossey-Bass.

Michigan Department of Education. (n.d.). Michigan K-12 Standards Mathematics.

https://www.michigan.gov/-/media/Project/Websites/mde/Literacy/Content-Standards/ELA_Standards.pdf?rev=0f76588bc2bd48f89165484fa35d2b31

Intervention: How will we respond when they don't learn?

- ☐ Differentiated Instruction
- ☐ Target specific skills
- ☐ Data item analysis
- ☐ Leveled materials (below, on level, above)
- ☐ Bloom's Taxonomy
- ☐ Grade recovery (re-do/correct)
- ☐ Parent contact

- ☐ Referral to Student Support Team
- ☐ Graphic organizers
- ☐ Manipulatives
- ☐ Choice boards
- ☐ Immediate feedback
- ☐ Flexible grouping
- ☐ Extended responses (math/reading)
- ☐ Journal/Reading logs

Intervention: What will we do if they don't learn it? (see guide)

- ☒ **Manipulatives** – Students who have not mastered the concept will use physical number line strips and integer tiles to build concrete understanding of positive and negative values.
- ☒ **Graphic organizers** – A structured notes template will be provided that includes a number line, a Frayer model for absolute value, and sentence stems for writing inequality statements.

Responses to Intervention (RtI)

- ☐ Small group instruction
- ☐ Tiered group instruction (Tier I, II, III)
- ☐ 1-1
- ☐ Centers (leveled)
- ☐ Re-teach in a different way
- ☐ Modify: backtrack, build background knowledge
- ☐ Tutoring: after or before school, lunch
- ☐ Referral to Student Support Team

- ☒ **Small group instruction** – The teacher will pull a small group during the enrichment/independent practice portion of the lesson to re-teach the concept of absolute value using a different approach, which will be distance from zero using a classroom floor number line.

☒ **Re-teach in a different way** – Instead of abstract notation, students will use color-coded cards, red for negative, blue for positive, to physically order numbers before transitioning to the number line diagram.

Enrichment: How will we respond if they already know it?

☒ **Centers – High level** – Students who demonstrate mastery early will move to an enrichment center where they solve real-world coordinate plane problems, such as plotting and finding distances between cities on a map grid.

☒ **Independent projects** – Advanced students will create their own illustrated real-world rational number scenario book that shows at least five examples of positive/negative numbers in context, such as temperature, elevation, finance, sports yardage, sea depth.

☒ **Peer tutoring** – Mastery students may serve as Math Mentors and assist a partner at the same table during guided practice, explaining their reasoning using academic vocabulary.

☒ **Games** – Students may engage with a rational number ordering card game, racing to correctly sequence a hand of rational number cards on their personal number line.

- ☐ Choice boards
- ☐ Use vocabulary to write sentences
- ☐ Accelerated reader
- ☐ Centers-High level
- ☐ Reading buddies
- ☐ Peer tutoring
- ☐ Enriched-Leveled Reader-Novels
- ☐ Picture/writing journals
- ☐ Independent projects
- ☐ Separate curriculum
- ☐ Games
- ☐ Group leader